Group 111

Simulating an energy-efficient In-Memory Compute AI accelerator by mapping Binarized Neural Networks to Spintronic MTJ arrays

Milestone 1: First Report

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Git Repository	https://github.com/mayazelkowicz-source/Spintronics_Simulator
Notion Workspace	https://app.notion.com/p/Spintronics-Lab-Simulating-an-energy-efficient-In-Memory-Compute-AI-accelerator-ce5240c4d77d83e7a51d818497730381

1. Problem Definition

Running Artificial Intelligence (AI) on battery-powered edge devices is limited by the "Von Neumann bottleneck": moving data between memory and processor dominates energy and latency. Spintronic memories built from Magnetic Tunnel Junctions (MTJs) can compute inside memory, but thermal noise and switching errors may corrupt results. Chip architects and device physicists lack a fast way to predict this impact.

Can AI inference stay accurate on noisy magnetic memory?

2. Background and Related Work

Background. *Spintronics* uses the electron's spin as well as its charge. Its key device, the **Magnetic Tunnel Junction (MTJ)**, places a thin insulator between two ferromagnetic layers: one fixed and one free to flip. Parallel (P) magnetizations give low resistance, Anti-Parallel (AP) high resistance (Tunnel Magnetoresistance) [1]. The free layer follows the **Landau–Lifshitz–Gilbert (LLG)** equation: the magnetization precesses around the effective field and is damped toward it [2]. A spin-polarized current exerts **Spin-Transfer Torque (STT)**, which opposes damping and can switch the bit [3]; thermal fluctuations make this probabilistic. A **Binarized Neural Network (BNN)** restricts weights and activations to ±1, which matches the two states of an MTJ.

Related work.

- **BinaryNet and XNOR-Net** [4, 5] showed ±1 networks keep much of full-precision accuracy with far less computation. Both assume ideal digital hardware.
- **Samsung MRAM crossbar** [6]: a 64×64 MTJ array classified handwritten digits with 93.23% accuracy (software baseline: 95.24%). Because MTJ resistance is low, summing currents on a column draws large currents, so the authors summed resistances instead.
- **Hirtzlin et al.** [7] simulated BNNs on spin-torque MRAM by injecting weight errors at given bit-error rates. Rates up to 0.1% had little effect on accuracy, allowing cheaper cells without error correction.
- **NeuroSim** [8] and **IBM AIHWKit** [9] are general analog-crossbar simulators, mainly for resistive and phase-change memories.

Our position. Like [7], our behavioral simulator represents device physics as statistical errors and noise injected during inference. We take the error rates and noise levels from LLG/STT models and HUJI Spintronics Lab measurements, adding read noise and resistance variation. We will also analyze how current and resistance summation [6] scale with array size.

3. Proposed Solution

Overview

Abstract view. How much of a stored BNN's accuracy survives realistic device physics, and at what energy cost? An MTJ is a nanoscale magnetic switch whose resistance stores one bit. In a crossbar, inputs drive row wires (*word lines*) and each column wire (*bit line*) sums the currents, computing a weighted sum in memory.

	Alt. 1: Fabricate hardware	**Alt. 2: Circuit simulation (SPICE, Simulation Program with Integrated Circuit Emphasis)**	**Alt. 3: Architectural simulator (chosen)**
Idea	Build and measure a real MTJ array	Model every device and wire electrically	Behavioral arrays with statistical noise

Pros	Real measurements	Electrically precise	Fast; wide parameter sweeps
Cons	Slow, costly fabrication	Slow for our ~200,000 cells	Depends on calibration

Solution proposal

We choose **Alternative 3**: each point of a noise-parameter sweep needs inference on the full 10,000-image test set over many seeds, which is practical only with a fast behavioral model. Lab measurements and published circuit-level results [10] reduce calibration risk.

Key components

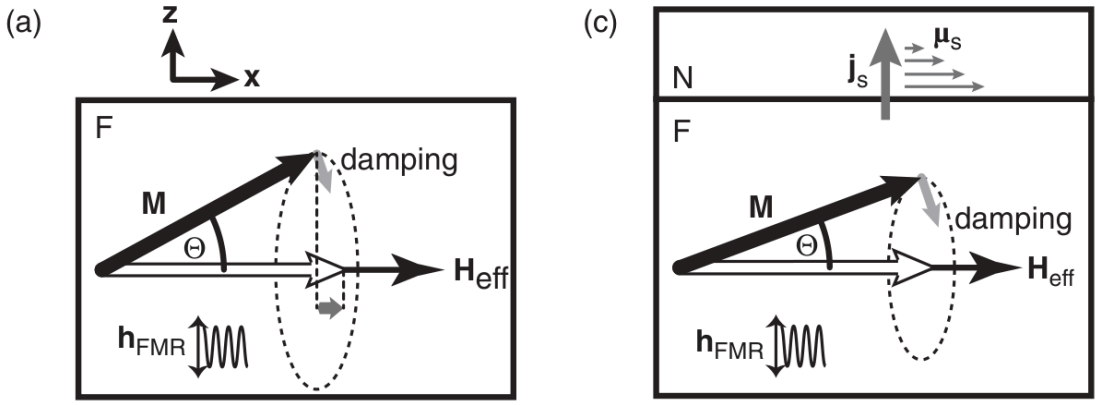


Figure 1 – Magnetization dynamics described by the LLG equation (from the course reading, Althammer et al. [2], Fig. 6.1a,c).

Figure 1 shows the physics behind the Noise Injector: in (a) the free layer precesses and is damped into a stable P or AP state storing a ±1 weight; (c) shows angular-momentum transfer, whose reciprocal (STT) writes weights. Thermal agitation can cause unintended flips.

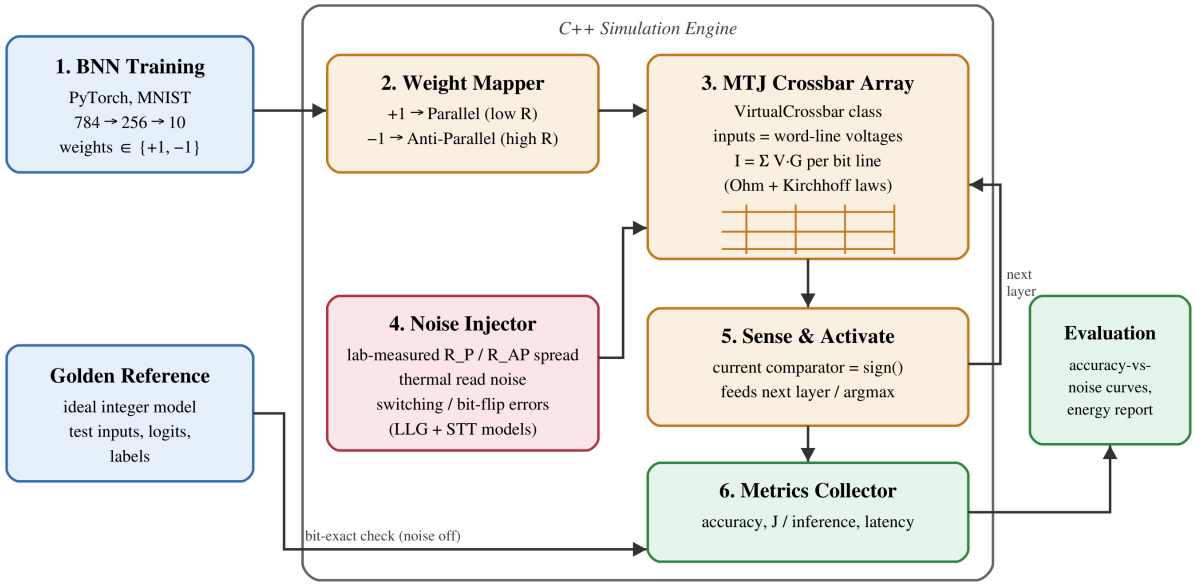


Figure 2 – Component diagram of the proposed simulator.

1. **BNN Training:** ±1 weights and *golden reference* data (trusted inputs, outputs, labels).
2. **Weight Mapper:** maps +1 to P and −1 to AP. A reference column or P/AP differential pairs remove the input-dependent current offset, so the comparator recovers the signed sum.
3. **MTJ Crossbar (C++ `VirtualCrossbar`):** inputs drive word lines; bit lines sum currents.
4. **Noise Injector:** resistance variation, read noise, physics-based bit flips.

5. **Sense & Activate:** sign() between layers; argmax over the ten outputs gives the predicted digit.
6. **Metrics Collector:** accuracy, energy, latency.

Solution feasibility

- **Algorithm:** our PyTorch BNN (784-256-10) reaches **94.30%** test accuracy on MNIST (Modified National Institute of Standards and Technology handwritten digits).
- **Simulator core:** C++ ideal inference reproduces the golden *logits* (raw per-class scores) exactly on all 10,000 test images.
- **Prototyping:** Python and C++ exchange CSV files, so retrained models need no code changes.
- **Modularity:** the code is split into loading, crossbar and activation parts; noise models will be added behind a common interface.

Technologies and skills

- **C++17, CMake** (https://cmake.org): simulation engine. *Skills:* efficient C++, Monte Carlo simulation.
- **PyTorch** (https://pytorch.org): BNN training. *Skills:* quantization-aware training.
- **Git, GitHub:** version control. *Skills:* branching, code review.
- **Notion:** project management.
- **Hardware knowledge:** MTJ physics, crossbar circuits, sense amplifiers, energy estimation.

Challenges and uncertainties

- **IR drop** (voltage loss along wires) distorts bit-line currents [10], worse with low MTJ resistance [6]. *Backup:* 64×64 tiles with first-order attenuation, or resistance summation [6].
- **Read disturb:** read current exerts STT and may flip cells. *Backup:* per-read flip probability.
- **Lab data availability.** *Backup:* published parameters [1, 3, 7].

Required knowledge

- **Literature:** Cullity (magnetic materials); Gurevich and Melkov (magnetization dynamics); Fabian, Žutić et al. (spin transport); [2, 3, 7]; Sze et al. (2017).
- **Courses:** Dr. Amir Capua's Spintronics reading course (LLG dynamics, spin currents, STT); Computer Architecture; Machine Learning; MIT 6.5930 *Hardware Architecture for Deep Learning* (online).

4. Evaluation and Verification Scheme

Persona. A device researcher at the HUJI Spintronics Lab asking: *"Is our measured MTJ stack good enough for AI inference, and which parameter must improve?"*

Verification (correctness).

- With noise disabled, the integer model must match the golden logits exactly on the full **10,000-image** MNIST test set (achieved), and the conductance model must predict the same class for every image.
- Unit tests check each physical model against analytical limits (e.g., zero temperature gives zero flips).

Evaluation (quantitative targets).

- **Robustness:** accuracy drop **below 5 percentage points** relative to the ideal BNN under lab-calibrated or published-parameter noise.
- **Sensitivity sweeps:** bit-error rate 0–10% and resistance variation 0–20%; each point averaged over **at least 30 seeds**, with 95% confidence intervals.
- **Energy:** Joules per inference from cell read energy ($V^2 \cdot G \cdot t$) plus sensing overhead, compared with a Von Neumann baseline from published operation and memory-access energies [11]. Projected target: at least 10× lower, to be confirmed by our model.
- **Latency:** crossbar read cycles per inference versus sequential multiply-accumulate operations.
- **Simulator speed:** full test set in **under 60 seconds** on a laptop.

Real-world testing. We will calibrate the Noise Injector with lab measurements (resistance spread, damping) and test degraded inputs (noisy, rotated digits; Fashion-MNIST).

5. Annual Work Plan

First steps, November–December 2026 (essential):

- Golden-reference verification on all 10,000 test images.
- MTJ resistance mapping and bit-line current summation.
- Collecting device parameters from the Spintronics Lab.

MVP (Minimum Viable Product), "Skateboard", Demo Day January 2027 (essential): our existing C++ core extended with an MTJ conductance model and a statistical bit-flip Noise Injector, producing an accuracy-versus-noise curve for the MNIST BNN.

"Bike", March–April 2027 (essential):

- Physics-derived noise: thermal stability, STT switching probability, read disturb.
- Energy and latency models.
- *Nice-to-have:* tiling with an IR-drop approximation; analytical comparison of current and resistance summation.

"Car", May–June 2027:

- *Essential:* final evaluation against the targets in Section 4; basic device-requirement map from lab or published parameters.
- *Nice-to-have:* lab-calibrated device-requirement map; Fashion-MNIST; comparison with NeuroSim; noise-aware retraining.

Responsibilities: Maya Zelkowicz leads physical and noise modeling and machine learning; Sarit Mishiev leads C++ engine architecture and optimization; evaluation is joint.

6. Project Management

- **Notion** (https://app.notion.com/p/Spintronics-Lab-Simulating-an-energy-efficient-In-Memory-Compute-AI-accelerator-ce5240c4d77d83e7a51d818497730381): project timeline, task assignments per member, requirement tracking and weekly progress updates.
- **Git** (https://github.com/mayazelkowicz-source/Spintronics_Simulator): all code is versioned on GitHub. Each member commits their own work with descriptive messages, and the README documents setup and run instructions. Regression testing is manual: before

pushing, we run the C++ test and confirm exact agreement with the golden reference.

References

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